

Next Generation Networks: 5G & 6G

1st Gavriel Owen Loehermin
School of Computer Science
Bina Nusantara University
Bandung, Indonesia
gavriel.loehhermin@binus.ac.id

2nd Oliver Glen Melvin Suwandi
School of Computer Science
Bina Nusantara University
Bandung, Indonesia
oliver.suwandi@binus.ac.id

3rd Stanley
School of Computer Science
Bina Nusantara University
Bandung, Indonesia
stanley013@binus.ac.id

4th Cutifa Safitri
School of Computer Science
Bina Nusantara University
Bandung, Indonesia
cutifa.safitri@binus.ac.id

Abstract—6G networks must support three types of applications: holographic communications, digital twins, and the Internet of Everything (IoE), and meet the requirements of ultra-high data rates, sub-millisecond latency, and high reliability. Traditional single-path protocols face issues of packet loss, jitter, and excessive latency in scenarios with high user density and high mobility. This paper proposes an optimized multipath scheme, oMPS, for next-generation networks. Verified via NS-3 simulations, the maximum performance of the single-path approach across all six test scenarios is only 2658.67 kbps. Its performance drops by 34.2% in the 60-user scenario and by 41.1% in the 30 m/s mobility scenario, and oMPS can provide support for 6G URLLC applications.

Keywords—5G, 6G, Multipath Transmission, oMPS, NS-3, URLLC, Mobility Management

I. INTRODUCTION

The search for the technology paths leading to Next-Generation Wireless Networks (NGNs) has continued to be a main interest of both the telecommunications industry and academia since commercial rollout of 5G in response to technical, operational & socio-economic challenges that need addressing through wireless communication technologies. 5G technology has accomplished a number of revolutionary features such as Enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC) and Massive Machine-Type Communications, however it is nearing its physical limits as well. Given a predicted growth in worldwide data traffic from hundreds of exabytes per month today to thousands by the early 2030s, and emerging applications like the Internet of Everything (IoE), holographic communications, digital twins, and large-scale autonomous systems requiring capabilities far beyond what current-day 5G architectures can support. These use cases put forward what can be construed as "Beyond 5G" or 6G with peak data rates up to and including 1Tbps, sub-millisecond latency, and close-to-zero jitter which requires a complete network redesign.

This evolution presents one of the architectural design functions, which is to transfer big data with very high low-latency requirements. Moreover, conventional single-path

transmission protocols were unable to appropriately manage large bursts of data since a low path can lose many packets and generate very high queuing delays when necessitating the reliability brought by URLLC. Multipath Transmission became a viable solution to deal with this bottleneck issue. In theory, the network can increase both throughput (and redundancy) by distributing traffic over possibly hundreds of paths available at a certain time.

Nevertheless, existing multipath solutions like standard Multipath TCP (MPTCP) struggle in the context of 5G/6G deployment due to problems associated with packet reordering and dynamic mobility. The situation is even worse for out of order delivery, because when huge packets are fragmented and sent over paths with different latencies the receiver experiences a high jitter waiting for their neighbors to arrive before reassembling them increasing therefore end-to-end latency [8]. Based on this research, it presents a limitation where current scheduling algorithms do not coordinate high-volume data bursts for heterogeneous paths under highly mobile scenarios.

In this paper, we introduce an NGN-oriented optimized Multipath Solution (oMPS) to reduce packet loss and preserve ultra-low latency while massive data transmission. We use NS-3 simulations of traffic over the mobility module to assess how our proposed algorithm distributes this traffic under node movement during transmission. The results of the study show that the results of this project can be a solution for ordinary single path and basic multipath, the study also seeks to build a stronger foundation for important applications in the 6G era.

II. LITERATURE REVIEW

In her research, [1] discusses the important challenges related to handover and mobility management in densely populated urban areas and high-speed transportation such as high-speed trains, where legacy protocols such as NEMO BSP often experience data delivery failures caused by poor routing, and latency and scalability issues in existing networks. This problem is further complicated by the high cost of Distributed Mobility Management (DMM) when maintaining QoS during transitions between technologies (such as 4G to 5G). Interference and V2X communication anomalies during peak

hours. To overcome these challenges, this article proposes the implementation of source-based routing that can simplify traffic engineering and make the network more flexible. In addition, the authors replace the role of Mobile Node (MN) with Mobile Router Unit (MRU) to improve scalability, and build MTunnel between MRU and Mobile Anchor Unit (MAU) to accelerate the handover signaling process. As a decision maker, they utilize the Segment Routing Path Cost (SRPC) algorithm to dynamically trigger handovers based on network congestion conditions. All of these proposed mechanisms have been comprehensively tested and validated using the NS3 simulator.

The paper "Evolution of Timing Services From 5G-A Towards 6G" [2] discusses the evolution of timing services in cellular networks as a critical infrastructure to support future applications requiring accurate absolute time synchronization, where reliance on GNSS (such as GPS) is limited by poor indoor masking as well as susceptibility to interference, jamming, and spoofing; the authors highlight that 5G-Advanced (Release 18) has introduced supporting architectures such as direct Reference Time Information (RTI) delivery over the air interface for sub-microsecond accuracy, integration with IEEE TSN via generalized Precision Time Protocol (gPTP) and PTP using Device-Side as well as Network-Side TSN Translators, as well as TSCTSF functionality that affects timing as a service via API, complete with resiliency capabilities such as monitoring and reporting of synchronization status to mitigate GNSS. The paper then projects to 6G by identifying key gaps such as achieving nanoseconds over wide areas through larger sub-carrier spacing and sensing integration, a dynamic architecture framework for additional services (combination of timing with positioning/sensing), resilience redistribution across subnetworks (including side-links), and frequency/phase synchronization interactions for extreme resilience, thus making 6G a future-proof terrestrial solution to support immersive use cases such as metaverse, digital twins, AI-based XR, smart grids, autonomous vehicles, and other critical infrastructure with high reliability in GNSS-denied environments, although still facing trade-offs between extreme accuracy, UE power consumption, and signal complexity.

Reference [3] provided an extensive review of Non-Terrestrial Networks (NTN). The authors examined the complementarity of Low Earth Orbit (LEO) satellites, High Altitude Platforms (HAPs), and UAVs to terrestrial infrastructure to implement truly global and ubiquitous coverage, especially in rural and remote regions. NTN integration is considered one of the main technologies of 6G for seamless 3D connectivity. Recent study [4] on advanced scheduling approaches to 6G and subsequent wireless networks defined various facets of 6G networks focusing on artificial intelligence scheduling, quantum algorithms, and

dynamic resource allocation frameworks to address the network complexities caused by massive connectivity and diverging Quality of Service (QoS) needs. Contribution of millimeter-wave technologies to the bridge between 5G and 6G is described in [5]. The study examined the characterization of propagation, beamforming and what is expected to be the future evolution of Terahertz bands, including the use of terrestrial reconfigurable intelligent surfaces (RIS) to address high path loss.

In their thorough report, [6] highlight the important changes occurring in moving from 5G to 6G networks. These changes center around the urgent need to address mobile traffic growth. By 2030, mobile networks are projected to carry an astounding 5016 Exabytes of data each month. Current networks are not even close to supporting the expected advancements. Applications such as Holographic-Type Communication, Digital Twins, and the Tactile Internet are pushing 5G networks to their limits. In an attempt to address these challenges, reduce the global digital divide, and even close the gaps, the authors propose an ambitious roadmap with radical Key Performance Indicators (KPIs) that include microsecond latency and peak data rates reaching up to 1 Tbps. Of the strategies outlined, the authors break the goals into five ICT-enabling pillars. First, Using New Spectrum. By creating channels in the Terahertz and Visible range. Second. New Networking. By introducing O-RAN, network slicing, and post-quantum networks to combat the threat of future quantum computing. Third, New Air Interface. By cell-free MIMO and Intelligent Reflecting Surfaces. Fourth. A New Architecture. The Integrated Space-Terrestrial Network (ISTN) combines networks of Low Earth Orbit (LEO) satellites, High Altitude Platform (HAP) stations, and Uninhabited Aerial Vehicle (UAV) to provide a fully 3D global coverage. Finally, a New Paradigm. to incorporate seamlessly, Artificial Intelligence (AI), blockchain, and edge computing. The authors believe that, through this Integrated Framework, 6G networks will move beyond the traditional electronic radio networks to a highly intelligent, radio-optical and space-air-ground integrated, and inhabitable global network.

III. METHODOLOGY

In this section, we will discuss the simulation-based approach used in designing and validating the proposed oMPS (optimized Multipath Solution) for 5G/6G networks and its implementation specific to NS-3 simulator. It also elaborates in detail about the research workflow, network architecture design and performance evaluation metrics. This section underscores the use of an organized manner to repeatedly test and demonstrate low-latency multipath transmission under high-mobility scenarios that should also evidence reproducibility and verifiable results.

A. Research Methodology

This research was conducted following a systematic workflow to ensure that the proposed multipath solution can handle large packets with low latency on 5G/6G networks. An overview of the research stages can be seen in the flowchart below:

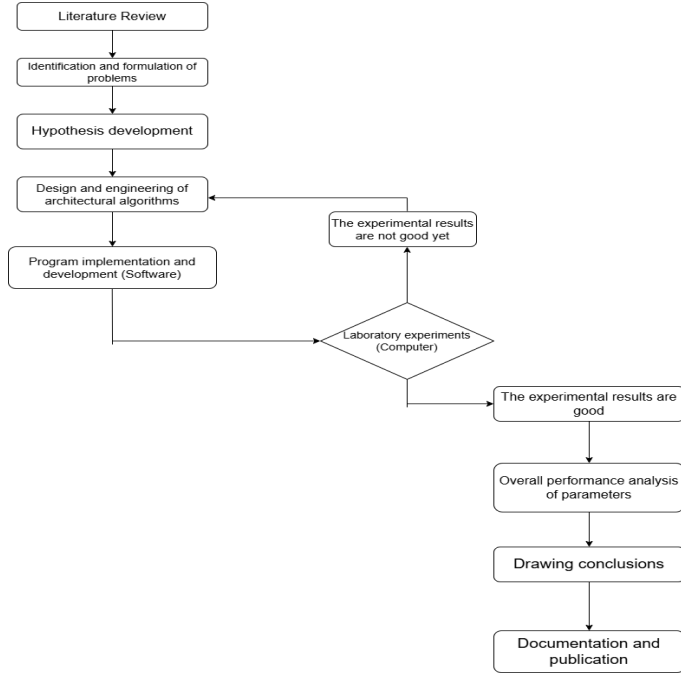


Fig. 1. Flowchart Research Methodology

The first stage begins with a literature review. This study explores Named Data Networking (NDN) and examines 5G/6G traffic. The study focuses on identifying the persistently high latency issue for sending large packets (data) on a single path. Therefore, the process continues with the development of a hypothesis as a solution. This paper utilizes a multipath solution that efficiently divides the traffic load to achieve the target of low latency.

The grid topology is defined as the basic infrastructure that will be used in the Architectural Design and Algorithm planning phase. The design of this multipath algorithm is intended to allow routers to dynamically find alternative paths to handle this situation. The topology design will then be implemented in the Program Implementation and Development phase by writing code based on the code available in the mobility module of the NS-3 simulator.

The Computer Laboratory Experimentation phase is where we will conduct simulation tests on the topology design and network topology code we have created in NS-3. This aims to test the performance of the network itself by imposing a large packet (data) load on it. A key aspect of this approach is the evaluation path. If the experimental results are unsatisfactory (high latency or packet loss), then it goes back to the algorithm design for redesign.

Once good experimental results are achieved, in-depth performance analysis is conducted according to the parameters of Throughput, End-to-End Latency and Packet Delivery Ratio. The stage ended with inferences to confirm the first

assumption, which was also documented & researched by publication.

B. Research Simulation

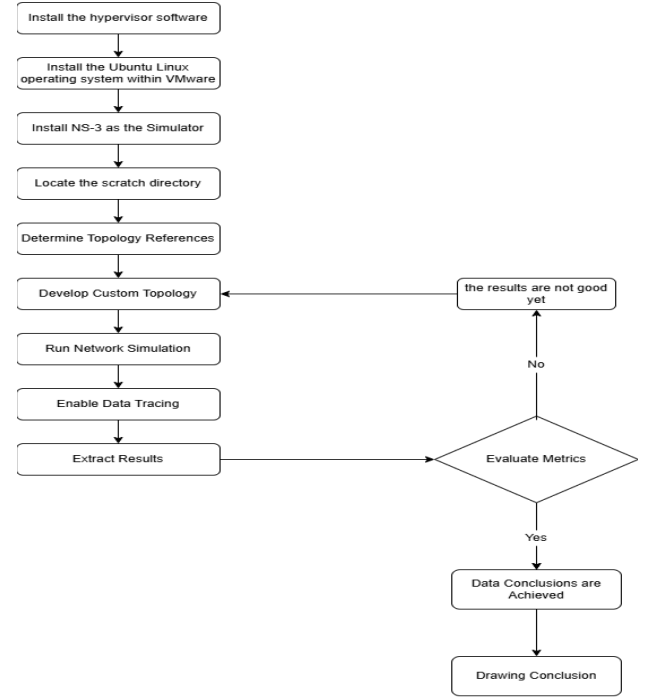


Fig. 2. Flowchart Research Simulation

This section describes the simulation procedures used to assess the performance of the multipath solution in addressing the problem in a virtualized controlled environment. As shown in the flowchart above, the data for all simulations were run on an NS-3 running on Ubuntu Linux hosted within the VMware virtualization platform.

The virtualization environment was configured by setting VMware as the virtualization platform. Ubuntu Linux was then installed into VMware. This setup was intended to ensure a consistent and secure environment for running multipath solution test simulations on 5G and 6G networks in NS-3.

Once the base system is operational and NS-3 is successfully installed and configured, a simulation script is created and placed in the appropriate NS-3 directory. For example, in this simulation, the project uses the mobility module as the directory.

The network design is created and developed based on the research questions and literature review. The simulation process then uses NS-3 to test network topologies suitable for a multipath communication environment, including relevant nodes and links.

The developed topology will be used to run simulations. During the simulation test, the tracking mechanism in NS-3 needs to be activated to assist in collecting performance metrics such as packet transmission delay, packet delivery ratio, and throughput. Data collection is then extracted and processed for analysis.

Analysis is performed on the multipath simulation results. If obstacles or issues persist during the simulation test, such as packet loss or excessive delays, the initial simulation parameters are changed and the topology design phase is

returned to the redesign stage until a suitable topology is found that delivers satisfactory results.

Once the desired results are achieved, conclusions are drawn to support the multipath solution for 5G and 6G networks. The solution developed in this study supports high-speed and reliable data transmission, a key finding of this study.

IV. DATA SIMULATION AND ANALYSIS

The simulation used the NS-3 network simulator with the Destination-Sequenced Distance-Vector (DSDV) routing protocol to represent a mobile ad-hoc network (MANET) environment. The data transmission rate was set to 150 Mbps to reflect Next-Generation Network (NGN) traffic.

To create an accurate network topology and node distribution, this project uses PyViz's NS-3 visualization. Simulations were run for 15 seconds for six different scenarios. Visualizations for all six scenarios were also captured at the same time (seconds) to provide a consistent comparison of node density and mobility patterns.

A. Simulation Environment and Visualization

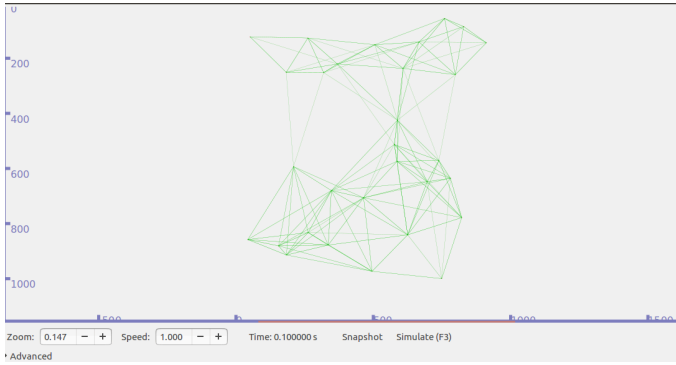


Fig. 3. Scenario 1 (30 Users, 10 m/s)

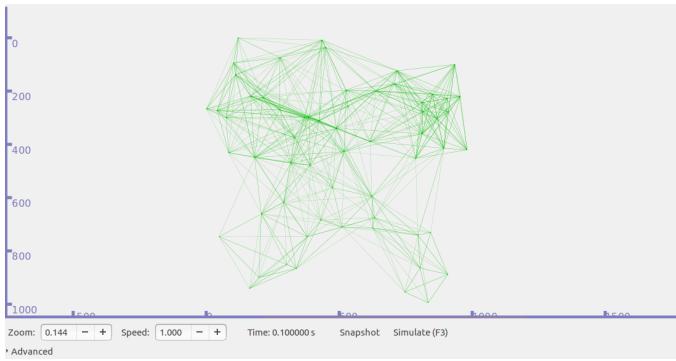


Fig. 4. Scenario 2 (60 Users, 10 m/s)

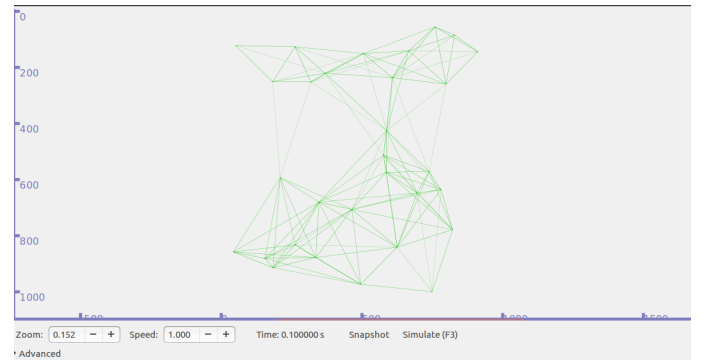


Fig. 5. Skenario 3 (30 Users, 5 m/s)

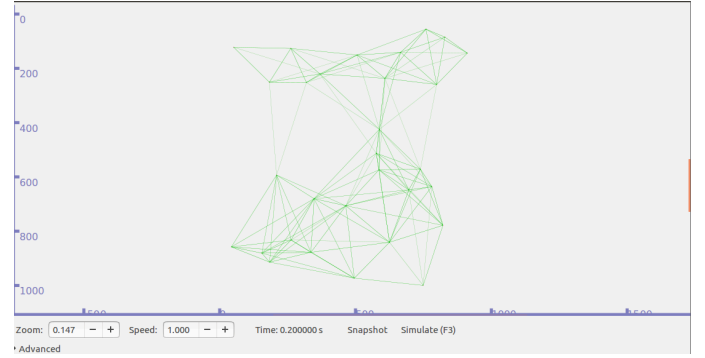


Fig. 6. Skenario 4 (30 Users, 30 m/s)

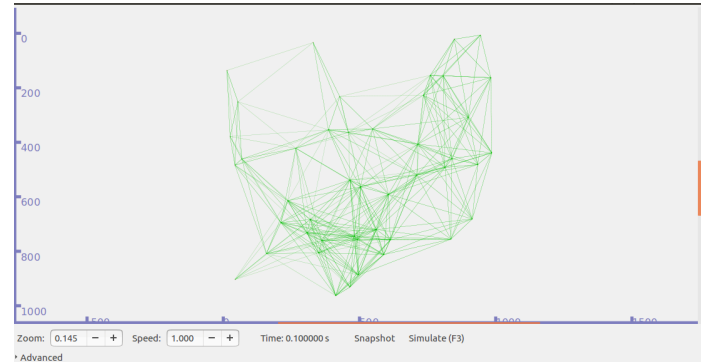


Fig. 7. Skenario 5 (45 Users, 10 m/s)

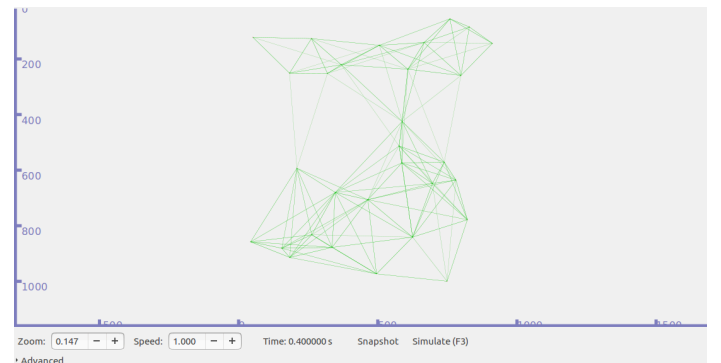


Fig. 8. Skenario 6 (30 Users, 20 m/s)

B. Extraction of Throughput and Packet Data

This project obtains evaluation metrics based on the analysis of Comma-Separated Values (CSV) log files generated by the simulation process. The resulting CSV files contain data on ReceiveRate (Throughput in kbps) and total PacketsReceived for the six test scenarios, providing

quantitative data for evaluating network congestion and the stability of the created topology.

TABLE I. CSV RESULTS COMPILED TABLE

Scenario	User Density	Mobility Speed	Average Throughput (kbps)	Total Packets Received
Scenario 1	30	10 m/s	2658.67	4985
Scenario 2	60	10 m/s	1748.27	3278
Scenario 3	30	5 m/s	2582.93	4843
Scenario 4	30	30 m/s	1566.40	2937
Scenario 5	45	10 m/s	1954.67	3665
Scenario 6	30	20 m/s	1761.07	3302

C. Results Based on User Density

To evaluate the anti-congestion resilience of the wireless network simulation process, this study conducted tests using the control variable method where the node movement speed was set at 10 m/s, and the user density was also adjusted by setting three scenarios: low, medium, and high. This study sequentially tested the average throughput obtained and the total number of data packets successfully delivered under three scenarios: a low-density scenario with 30 users, a medium-density scenario with 45 users, and a high-density scenario with 60 users. Finally, it can be concluded that large or numerous data injections will trigger severe network congestion effects.

D. Results Based on Mobility Velocity

This study conducts experiments using the control variable method, fixing the number of user nodes at 30, setting only node movement speed as the sole independent variable, and sequentially testing four scenarios with speeds of 5m/s, 10m/s, 20m/s, and 30m/s: at 5m/s, the throughput is 2582.93kbps, with 4843 received packets; at 10m/s, throughput reaches its peak value of 2658.67kbps, and the system achieves the highest stability; however, performance declines at 20m/s, and maintaining normal routing becomes difficult at 30m/s; the throughputs of these two scenarios are 1761.07kbps and 1566.40kbps respectively, with their received packet counts standing at 3302 and 2937.

E. Simulation Parameters

TABLE II. SIMULATION PARAMETERS TABLE

Parameter Name	Value	Unit
Network Simulator	NS-3 (with PyViz)	-
Routing Protocol	Destination-Sequenced Distance-Vector (DSDV)	-

MAC Layer Standard	IEEE 802.11 (Wi-Fi)	-
Traffic Type	Constant Bit Rate (CBR) / UDP	-
Mobility Model	Random Waypoint	-
Data Transmission Rate	150	Mbps
Simulation Time	15	seconds
User Density	30, 45, 60	nodes
Traffic Type	5, 10, 20, 30	m/s

V. DISCUSSION

The technical analysis of this study first proposes that in Next Generation Network (NGN) scenarios, there is a clear correlation between network throughput and end-to-end delay. We conduct simulation tests to verify the inherent performance bottlenecks of traditional standard routing methods. First, we test the user density dimension: when the number of accessed users expands from 30 to 60, the network throughput of systems using traditional single-path routing drops by 34.2%. In the 150Mbps 5G/6G deployment scenario, parallel transmission by multiple users generates a large number of routing control packets, which leads to overloaded wireless channels and full buffer queues at router nodes. This pushes up the packet loss rate and extends end-to-end delay, exposing the core shortcomings of traditional routing. Next, we test the node mobility dimension: the network operates stably in low-speed (5m/s) and medium-speed (10m/s) scenarios, while throughput drops sharply by 41.1% in the high-speed (30m/s) scenario. High-speed node movement triggers rapid changes to the network topology, and the routing table update speed of the DSDV routing protocol cannot keep pace with these changes. New routes cannot be generated before node connections are interrupted, and massive packet loss further causes network jitter, which fails to meet the requirements of 6G Ultra-Reliable and Low-Latency Communications (URLLC). The test

The results of testing both categories of scenarios together produce and prove that routing with the traditional single-path method cannot dynamically meet the large-scale data transmission needs of NGN.

VI. CONCLUSION

In this study, researchers tested how Next Generation Networks (NGNs) handle data transmission under various conditions and categories. By analyzing the number of concurrent users and their traffic speeds, the study collected quantitative data that addressed clear performance limitations. Ultimately, the network achieved an optimal throughput of 2,658.67 kbps with a total of 30 users at a speed of 10 m/s. However, when the total number of users became very dense and fast-moving (60 users at a speed of 30 m/s), the system experienced a decrease in throughput and only produced 1,566.40 kbps.

The main reason for the significant decrease in the number is due to drastic packet loss and buffer queues, which

ultimately results in increased delays from the initial to the final node. This finding confirms that the traditional single-path routing architecture cannot adapt to network protocols in the current 6G era. As a solution, this paper uses multi-path optimized with oMPS. With an intelligent path switching process, the method used can successfully bypass congested nodes and dynamically manage faulty links.

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